

Sentiment Analysis of Amazon Product Reviews

NLP Capstone Project – spaCy and spaCyTextBlob

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1. Dataset description

The analysis uses the *Datafiniti Amazon Consumer Reviews of Amazon Products* dataset (May 2019 version) from Kaggle. It contains **28,332 reviews** and **24 columns**. Two columns are used: **reviews.text** (the review, which is the input to the sentiment model) and **reviews.rating** (a 1–5 star rating, used only as a reference label when evaluating the model). No values were missing in reviews.text.

The ratings are heavily skewed towards positive reviews: 90.2% of reviews have 4 or 5 stars, 5.6% have 1 or 2 stars and 4.3% have 3 stars. This imbalance matters when interpreting the evaluation.

Rating	1	2	3	4	5
Reviews	965	616	1,206	5,648	19,897

2. Preprocessing steps

- **Missing values:** rows with no review text were dropped with `dropna(subset=['reviews.text'])` (none were removed in this dataset).
- **Normalisation:** each review was converted to a string, lowercased and stripped of leading and trailing whitespace.
- **Tokenisation and stop word removal:** the text was tokenised with spaCy's `en_core_web_md` model. Stop words (`.is_stop`), punctuation and whitespace tokens were removed, and the remaining tokens were joined back into a string.
- **Scoring:** the cleaned text was passed through the spaCy pipeline with the `spaCyTextBlob` component, which gives a polarity score from -1 (very negative) to +1 (very positive). A review is labelled **Positive** if polarity is above 0.1, **Negative** if below -0.1, and **Neutral** otherwise. The 0.1 neutral band was chosen after comparing bands from 0.0 to 0.3 (Section 3.3).

Example: *"Bulk is always the less expensive way to go for products like these"* becomes *"bulk expensive way products like"*.

3. Evaluation of results

3.1 Spot checks on sample reviews

The model was tested on the first five dataset reviews (judged by reading them) and on six hand-written reviews of known sentiment. It was correct on 4 of 5 dataset reviews and 5 of 6 hand-written reviews. The errors were:

- *"Bulk is always the less expensive way to go..."* (actually positive) was scored **Negative** (-0.500). Removing the stop word "less" left "expensive", which TextBlob scores as negative. The unprocessed text also scored negative (-0.333), so the lexicon is the main cause.
- *"The product is okay, nothing special but does the job."* (expected Neutral) was scored **Positive** (0.429) because the words "okay" and "special" are scored positively and the model does not understand "nothing special".

Stop word removal changed polarity scores on 2 of the 5 dataset reviews and 3 of the 6 hand-written reviews, but never changed the resulting label in these samples. Five samples are too few to draw firm

conclusions, so these are best read as illustrations.

3.2 Evaluation against star ratings

To evaluate on more data, a random sample of 1,000 reviews (`random_state=42`) was labelled using the star rating (4–5 = Positive, 3 = Neutral, 1–2 = Negative) and compared with the model's predictions. The model reached an **accuracy of 74.3%** (743 of 1,000). However, always predicting "Positive" would score **90.0%** on the same sample, so overall accuracy is misleading here and the per-class results matter more.

Actual \ Predicted	Negative	Neutral	Positive	Recall
Negative (50)	8	17	25	16%
Neutral (50)	5	15	30	30%
Positive (900)	16	164	720	80%
Precision	28%	8%	93%	

Recall: of the reviews that truly are in a class, the share the model found. Precision: of the reviews the model put in a class, the share that truly belong. Average recall across the three classes (balanced recall) is 42.0%.

The model is reasonably reliable when it predicts Positive (93% precision) but misses most negative reviews: half of the negative reviews (25 of 50) were predicted Positive. Many positive reviews (164 of 900) were labelled Neutral, which makes the Neutral class very unreliable (8% precision).

3.3 Choosing the neutral band

With a band of 0.0, a review is Neutral only if its polarity is exactly zero. Wider bands were tested on the same 1,000 reviews:

Band	Accuracy	Negative recall	Neutral recall	Positive recall	Balanced recall
0.0	78.3%	24%	12%	85%	40.3%
0.05	76.7%	20%	18%	83%	40.4%
0.1 (used)	74.3%	16%	30%	80%	42.0%
0.2	69.1%	10%	50%	73%	44.5%
0.3	59.2%	4%	64%	62%	43.3%

A wider band raises Neutral recall but lowers accuracy and Negative recall at every step. A band of 0.1 was chosen as a compromise. Balanced recall is highest at 0.2, but that setting loses more accuracy and catches fewer negative reviews. The gains are modest: the neutral band does not fix the model's main weaknesses.

3.4 Review similarity

`spaCy's similarity()` was used to compare reviews. Two positive reviews about price scored 0.933, but a negative review compared with a positive one still scored 0.867. Similarity is based on averaged word vectors, so it measures how close the topics and vocabulary are, not whether the sentiment matches.

4. Strengths and limitations

Strengths

- Fast and simple: no training data or model training is needed, and the polarity score is easy to interpret.
- Works well on reviews with clear sentiment words such as "great", "terrible" or "happy" (80% recall and 93% precision for Positive).
- Modular pipeline: cleaning, scoring and labelling are separate functions, so each step can be changed independently.

- A neutral band improved Neutral recall from 12% to 30%.

Limitations

- **No context understanding:** the lexicon scores words individually, so phrases like "less expensive" or "nothing special" are misread.
- **Stop word removal can hurt:** spaCy's stop word list includes words such as "not", "no" and "less", which can remove negation and modifiers.
- **Weak on minority classes:** only 16% of negative reviews and 30% of neutral reviews were found. Accuracy (74.3%) is below the 90.0% majority-class baseline.
- **Evaluation caveats:** the sample contained only 50 negative and 50 neutral reviews, so these percentages are rough. The neutral band was tuned and evaluated on the same sample. Star ratings are also an imperfect proxy for sentiment, and treating 3 stars as Neutral is an assumption.
- **Similarity measures topic, not sentiment** (Section 3.4).

Possible improvements

- Keep negation words (for example "not") when cleaning, or score the unprocessed text.
- Train a classifier (such as logistic regression on TF-IDF features, or a transformer model) using star ratings as labels, with class weighting to handle the imbalance.
- Choose settings on one sample and confirm them on a separate held-out sample.